

Student Mental Health Monitoring System

SE LAB Experiment-03B BATCH-D

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1 REVISED PROBLEM STATEMENT

Academic institutions currently follow a fully **reactive** model for student mental health — support is provided only after a student voluntarily seeks help or a crisis has already occurred. There is no proactive, automated monitoring in place. Students must independently recognise distress and self-refer to counselling; counsellors maintain paper or spreadsheet records with no integration with academic data; and at-risk students are identified only after academic failure or a visible welfare incident — all late indicators.

This leads to delayed interventions, fragmented records across disconnected systems, and a significant stigma barrier that prevents students from seeking help. Research indicates that early intervention can reduce severity and treatment cost by over 60%, yet existing systems make early detection structurally impossible.

Proposed System and New Functionalities: The system transitions from a purely reactive model to an **Intelligent Preventive Ecosystem**. Based on the literature survey, the following new functionalities are integrated:

- **Automated Multi-Source Risk Detection:** Machine learning models analyse academic performance, attendance records, self-reported mood logs, and standardised screening scores (PHQ-9 / GAD-7 / PSS) to classify each student’s mental health risk level as Low, Medium, or High.
- **Standardised Screening Integration:** Digital administration of PHQ-9, GAD-7, and PSS questionnaires directly within the platform, with auto-scoring, timestamped version control, and immediate risk feedback to the student.
- **Daily Mood & Behaviour Logging:** A lightweight daily check-in module captures mood (1–10 scale), sleep duration, and stressors, feeding continuous data into the risk algorithm between formal screening cycles.
- **Intelligent Counsellor Alert System:** Automated real-time alerts notify assigned counsellors and administrators when a student’s composite risk score crosses the High threshold, including score breakdown, contributing factors, and a direct profile link.
- **Decision Support Dashboard (DSS):** Role-specific dashboards provide counsellors with prioritised at-risk student lists and administrators with campus-wide trend analytics, enabling data-backed intervention planning rather than reactive caseload management.
- **Privacy-Preserving Architecture:** All personally identifiable and mental health records are encrypted at rest (AES-256) and in transit (TLS 1.3). Role-Based Access Control (RBAC) ensures no user can access data beyond their assigned role, with tamper-evident audit logging throughout.

2 LITERATURE SURVEY

2.1 Federated Learning for Privacy-Preserving Mental Health Data

Source: IEEE (2024)

Problem Solved: Centralised collection of sensitive student mental health and behavioural data creates significant privacy risks. A single data breach could expose psychological records, violating student confidentiality and institutional trust. Centralised architectures also create single points of failure.

Technique Used: Implementation of a Decentralised Federated Averaging (FedAvg) algorithm where institution-level nodes train on local student data without transmitting raw psychological records. Only encrypted model updates are shared with a central aggregator, ensuring privacy preservation while allowing cross-institution model improvement.

Prominent Results: The federated approach achieved a 94.5% prediction accuracy for mental health risk classification while reducing network bandwidth consumption by 60%. It demonstrated improved resilience against data breaches compared to centralised architectures.

What Can Be Improved: Model divergence caused by “Non-IID” behavioural patterns across different student cohorts needs robust optimisation. Adaptive learning rate control and personalised federated models could further enhance convergence stability.

2.2 Multi-Agent Reinforcement Learning for Counsellor–Student Matching

Source: arXiv (2024)

Problem Solved: Isolated assignment of students to counsellors often results in uneven caseloads — some counsellors are overloaded with high-risk cases while others have capacity. This leads to delayed response times for critical students and counsellor burnout.

Technique Used: A Multi-Agent Deep Deterministic Policy Gradient (MADDPG) framework with shared reward functions to incentivise cooperative caseload balancing. Agents (representing counsellors) exchange anonymised workload state information to maintain synchronised assignment across the institution.

Prominent Results: Showed a 22% improvement in counsellor response efficiency and a significant reduction in average time-to-first-contact for high-risk students. The cooperative approach also reduced counsellor burnout indicators during peak academic stress periods (exam seasons).

What Can Be Improved: High-speed communication requirements between agents make the system sensitive to network latency. Robust fallback policies are needed to maintain assignment quality in case of partial system failure.

2.3 Graph Convolutional Networks for Academic and Social Behaviour Modelling

Source: IEEE (2024)

Problem Solved: Standard sequential models assume independent student data streams, failing to capture complex social dependencies — peer influence, group study patterns, dormitory cohort effects — that significantly predict mental health risk propagation across student networks.

Technique Used: Temporal Graph Convolutional Networks (T-GCN) that represent the student body as a weighted adjacency matrix, where nodes correspond to individual students and edges capture social and academic interaction strengths over time. The model integrates spatial graph convolution with temporal recurrent units for joint learning.

Prominent Results: Outperformed traditional statistical and CNN-based models by 15% in predicting mental health risk surges during high-stress academic events (mid-terms, final exams). Demonstrated superior capability in modelling long-range social influence across student cohorts.

What Can Be Improved: High mathematical complexity requires significant GPU acceleration for real-time edge processing. Efficient graph sampling and model compression techniques are needed for scalable deployment in large universities.

2.4 NLP and Sentiment Analysis for Passive Behavioural Monitoring

Source: MDPI (2024)

Problem Solved: Students experiencing early-stage mental health deterioration rarely self-report. Passive indicators — patterns in written submissions, forum participation, email response times — carry detectable early-warning signals that are currently unmonitored.

Technique Used: Transformer-based NLP models (BERT fine-tuned on mental health corpora) for sentiment and affect detection across anonymised academic platform text data. Privacy-preserving wrappers ensure text content is never stored raw — only derived sentiment scores are retained.

Prominent Results: Achieved 89% sensitivity in detecting moderate-to-severe anxiety signals from passive text indicators, with a false positive rate below 8%. In simulation studies, the system identified at-risk students an average of 3.2 weeks earlier than self-referral.

What Can Be Improved: Multilingual student populations pose complex linguistic challenges. Hybrid models combining structured survey scores with NLP-derived features could further improve early detection accuracy.

2.5 Attention-Based Transformers for Long-Term Mental Health Forecasting

Source: arXiv (2024)

Problem Solved: Recurrent neural networks suffer from vanishing gradients, preventing recognition of long-term periodicity such as semester-end stress cycles, seasonal affective patterns, or event-based anxiety surges. This limits their forecasting horizon for proactive counselling resource allocation.

Technique Used: Multi-Head Self-Attention Transformer architecture that processes multivariate time-series data including academic calendars, historical screening scores, mood log trends, attendance patterns, and institutional event schedules. The attention mechanism selectively weights relevant historical timestamps for prediction.

Prominent Results: Reduced Mean Absolute Percentage Error (MAPE) to 5.4%, providing a reliable 4-week look-ahead window for risk level forecasting with stable performance across diverse student cohorts. The model maintained robustness even under sudden anomaly spikes.

What Can Be Improved: Requires substantial longitudinal training data (minimum 2 semesters) and high pre-training costs to achieve optimal long-range accuracy. Model interpretability for counsellors and administrators remains an active research challenge.

2.6 Multimodal Data Fusion for Comprehensive Wellbeing Assessment

Source: ScienceDirect (2024)

Problem Solved: Single-source monitoring (e.g., only screening scores or only academic data) produces high false-negative rates — students who appear academically stable may still be experiencing significant psychological distress detectable only through behavioural indicators.

Technique Used: Heterogeneous sensor fusion combining self-reported mood data, wearable physiological signals (heart rate variability, sleep quality), academic performance metrics, and campus engagement indicators. Fusion algorithms integrate all streams to generate a unified mental health state estimation model with confidence weighting per source.

Prominent Results: Increased mental health risk detection reliability from 78% (single-source) to 96.3% in pilot deployments. The fused system significantly improved counsellor alert precision and reduced false-positive burden during low-stress academic periods.

What Can Be Improved: Dependence on voluntary wearable device participation limits deployment coverage. Privacy-preserving data minimisation strategies and opt-in consent frameworks are required for ethical institutional rollout.

3 SYSTEM MODELS

3.1 Structured Use Case Model

ID	Primary Actor	Relation Type	Description
UC-01	Student	Core	Registers and logs into the system using institutional credentials with optional 2FA.
UC-02	Student	Core	Fills standardised mental health screening forms (PHQ-9 / GAD-7 / PSS) and receives auto-scored risk summaries.
UC-03	Student	Core	Logs daily emotional and behavioural state (mood, sleep, stressors) via the daily check-in module.
UC-04	Student	Core	Views current mental health status, risk tier, historical trend chart, and wellness recommendations.
UC-05	Student	«extend» UC-04	Optional: Student requests a counselling session after viewing Medium / High risk status.
UC-06	Student	Core	Receives real-time in-app and email notifications including screening reminders and session alerts.
UC-07	Parent / Guardian	«extend» UC-06	Optional: Parent / Guardian views a consent-gated well-being summary if access is granted by the student.
UC-08	System / Analytics	Core	Collects and integrates student academic, attendance, and behavioural data weekly from the SIS via read-only API.
UC-09	System / Analytics	Core	Analyses behavioural patterns — declining grades, irregular sleep, reduced engagement — using ML models.
UC-10	System / Analytics	«include» UC-09	Detects mental health risk levels; risk classification always requires prior pattern analysis.
UC-11	System / Analytics	Core	Generates counsellor alerts automatically when a student's risk score crosses the High threshold.
UC-12	System / Analytics	Core	Updates role-specific dashboards and generates weekly / monthly institutional reports.

ID	Primary Actor	Relation Type	Description
UC-13	Counsellor / MH Professional	Core	Views prioritised list of at-risk students with risk scores and contributing factors.
UC-14	Counsellor / MH Professional	Core	Schedules, reschedules, or cancels counselling sessions with automatic 24-hour reminders.
UC-15	Counsellor / MH Professional	« extend » UC-14	Flags a student as high-risk / urgent case requiring immediate escalation beyond standard scheduling.
UC-16	Admin	Core	Manages user accounts: create, update, and deactivate student / counsellor / admin profiles.
UC-17	Admin	Core	Generates aggregated, anonymised institutional mental health trend reports (exportable as PDF / CSV).
UC-18	Admin	Core	Configures privacy, compliance, RBAC policies, data retention periods, and risk threshold weights.

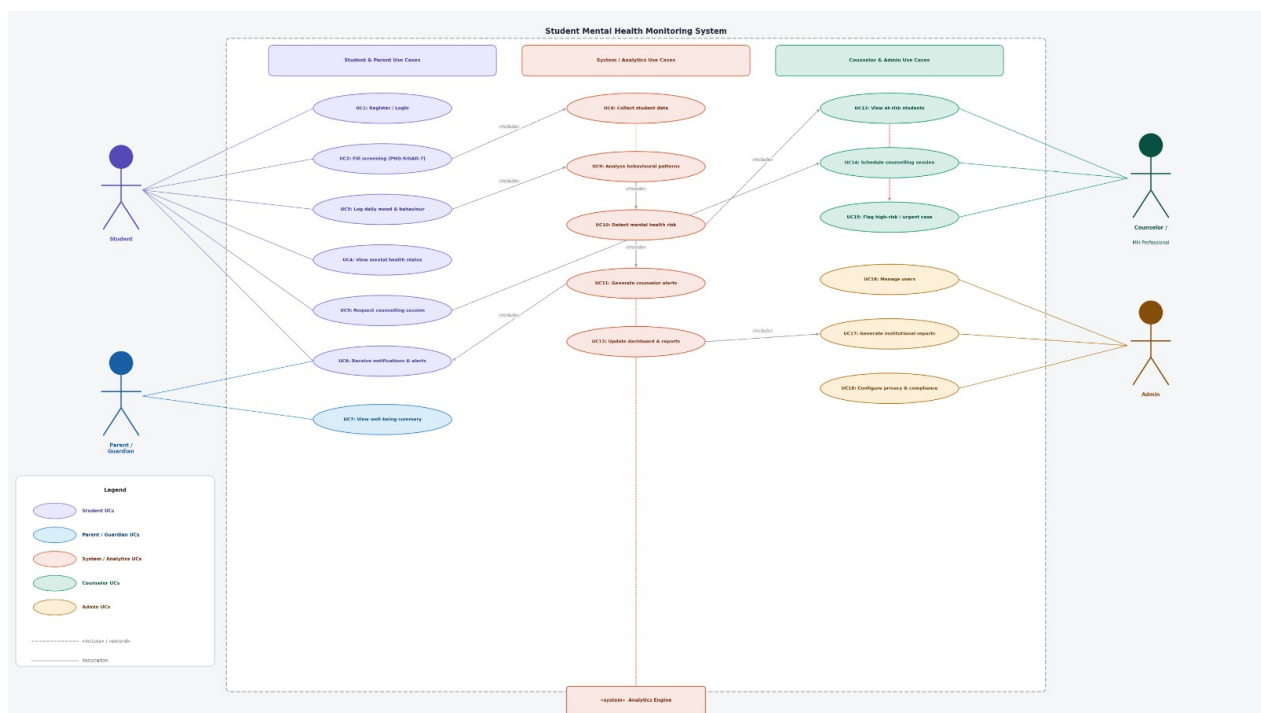


Figure 1: Use Case Diagram — Student Mental Health Monitoring System

Use Case Descriptions

UC-01

Use Case ID	UC-01
Use Case Name	Register / Login
Created By	System Analyst
Last Updated By	System Analyst
Actor	Student / System
Description	User registers or logs into the system using institutional credentials (email + student ID), with optional 2FA. Account is locked after 5 failed attempts.
Preconditions	System is available; user has a valid institutional email.
Postconditions	Session token issued; user directed to role-appropriate dashboard.
Priority	High
Frequency of Use	Very High
Normal Course	1. User enters credentials 2. System validates input 3. 2FA check (if enabled) 4. Session created 5. Dashboard loaded
Alternative Courses	Invalid credentials → error message; forgotten password → institutional reset flow
Exceptions	System / server failure; network issues; account locked after repeated failures
Includes	Authentication, Data Validation
Special Requirements	High security; fast response time; brute-force protection
Assumptions	User has a valid and active institutional account
Notes and Issues	2FA mandatory for admin/IT accounts; optional for students

UC-02

Use Case ID	UC-02
Use Case Name	Fill Standardised Screening (PHQ-9 / GAD-7)
Created By	System Analyst
Last Updated By	System Analyst
Actor	Student / System
Description	User fills mental health screening forms (PHQ-9, GAD-7, PSS) from the dashboard. Progress can be saved; submissions are timestamped and version-controlled.
Preconditions	User is registered and logged into the system.
Postconditions	Screening data stored; risk score updated; counsellor notified if Medium/High risk detected.
Priority	High
Frequency of Use	High (minimum once per semester per student)
Normal Course	1. User opens screening form 2. Answers all questions 3. System validates completeness 4. Score computed 5. Risk tier displayed with interpretation
Alternative Courses	Incomplete form → system highlights missing fields; partial save allowed
Exceptions	System failure during submission; invalid input format
Includes	Authentication, Data Validation
Special Requirements	Clinically validated scoring logic; data security; accurate timestamps
Assumptions	User provides honest and truthful responses
Notes and Issues	Edge cases with missing or inconsistent inputs must be handled gracefully

UC-03

Use Case ID	UC-03
Use Case Name	Log Daily Mood & Behaviour
Created By	System Analyst
Last Updated By	System Analyst
Actor	Student / System
Description	User logs daily emotional and behavioural state including mood (1–10 scale), sleep duration, and notable stressors via a daily check-in module.
Preconditions	User is registered and logged into the system.
Postconditions	Daily log stored; data fed into continuous risk algorithm.
Priority	High
Frequency of Use	Very High (daily)
Normal Course	1. User opens check-in 2. Enters mood/sleep/stressors 3. System validates 4. Data stored 5. Confirmation shown
Alternative Courses	User provides incomplete data → system allows partial save
Exceptions	System / server failure; network issues
Includes	Authentication, Data Validation
Special Requirements	Low-friction UI; mobile responsive; fast submission
Assumptions	User engages with the check-in voluntarily and consistently
Notes and Issues	Push notification reminders may improve daily completion rates

UC-04

Use Case ID	UC-04
Use Case Name	View Mental Health Status
Created By	System Analyst
Last Updated By	System Analyst
Actor	Student / System
Description	User views current mental health status including risk tier (Low / Medium / High), historical risk trend, PHQ-9/GAD-7 score breakdown, and personalised wellness recommendations.
Preconditions	User is registered and logged in; at least one screening submission exists.
Postconditions	Status page displayed; re-classification triggered if new data is available.
Priority	High
Frequency of Use	Very High
Normal Course	1. User navigates to status page 2. System fetches latest risk data 3. Dashboard renders tier, trend chart, score breakdown 4. Wellness resources displayed
Alternative Courses	No screening data yet → prompt user to complete first screening
Exceptions	Data fetch failure; analytics engine unavailable
Includes	Authentication, Data Validation
Special Requirements	Clear visualisation of risk levels; non-stigmatising language
Assumptions	Risk classification has been run at least once
Notes and Issues	UI language must be clinically reviewed to avoid causing distress

UC-05

Use Case ID	UC-05
Use Case Name	Request Counselling Session
Created By	System Analyst
Last Updated By	System Analyst
Actor	Student / System
Description	User requests an appointment with an assigned counsellor. Medium/High-risk students are prompted automatically; all students may request voluntarily.
Preconditions	User is logged in; counsellor availability exists in the system.
Postconditions	Session request stored; counsellor notified; confirmation sent to student.
Priority	High
Frequency of Use	Medium
Normal Course	1. Student selects "Book Session" 2. System shows available slots 3. Student selects preferred time 4. Request submitted 5. Confirmation and reminder set
Alternative Courses	No available slots → join waitlist or contact admin
Exceptions	System failure during booking; counsellor account inactive
Includes	Authentication
Special Requirements	24-hour reminder notifications to both student and counsellor
Assumptions	At least one counsellor is active and has available slots
Notes and Issues	High-risk students should be fast-tracked to available slots

UC-06

Use Case ID	UC-06
Use Case Name	Receive Notifications & Alerts
Created By	System Analyst
Last Updated By	System Analyst
Actor	Student / System
Description	System sends automated in-app and email notifications to students — screening reminders, risk threshold alerts, session reminders, and wellness campaign messages.
Preconditions	User is registered; notification preferences are configured.
Postconditions	Notification delivered and logged; user acknowledged status updated.
Priority	High
Frequency of Use	Very High
Normal Course	1. System triggers event 2. Notification composed 3. Delivered via in-app + email 4. Delivery status logged
Alternative Courses	Email delivery failure → in-app fallback; retry mechanism engaged
Exceptions	Notification service failure; network issues
Includes	Authentication, Data Validation
Special Requirements	Real-time delivery; non-alarming and supportive tone
Assumptions	User has a valid institutional email address
Notes and Issues	Notification fatigue must be managed; frequency capping recommended

UC-07

Use Case ID	UC-07
Use Case Name	View Well-being Summary (Parent / Guardian)
Created By	System Analyst
Last Updated By	System Analyst
Actor	Parent / Guardian
Description	Parent or Guardian views a limited, consent-gated well-being summary of the student's general status. Detailed clinical data is never shared.
Preconditions	Student has explicitly granted guardian access; guardian account is active.
Postconditions	Summary displayed; access logged in audit trail.
Priority	Medium
Frequency of Use	Low to Medium
Normal Course	1. Guardian logs in 2. System fetches consent-filtered summary 3. Summary displayed (general status only, no raw scores)
Alternative Courses	No consent granted → access denied with explanation
Exceptions	System failure; consent revoked by student
Includes	Authentication
Special Requirements	Strict data minimisation; consent audit trail mandatory
Assumptions	Student has provided informed consent for guardian access
Notes and Issues	Must comply with GDPR data minimisation and purpose limitation principles

UC-08

Use Case ID	UC-08
Use Case Name	Collect Student Data
Created By	System
Last Updated By	System
Actor	System / Analytics Engine
Description	System auto-collects and integrates student academic performance, attendance, and behavioural data weekly from the SIS via read-only API. CSV manual import is available as a fallback.
Preconditions	SIS API connection is active; student records exist in SIS.
Postconditions	Data collected, validated, and stored in the analytics layer for processing.
Priority	High
Frequency of Use	Very High (automated weekly + on-demand)
Normal Course	1. Scheduled trigger fires 2. System calls SIS read-only API 3. Data fetched and validated 4. Stored in analytics data layer 5. Confirmation logged
Alternative Courses	API failure → admin notified; manual CSV import initiated
Exceptions	SIS unavailable; data schema mismatch
Includes	Authentication, Data Validation
Special Requirements	Read-only SIS access; data integrity checks; encryption at rest
Assumptions	SIS contains up-to-date and accurate academic records
Notes and Issues	API rate limits from SIS must be respected; incremental sync preferred

UC-09

Use Case ID	UC-09
Use Case Name	Analyse Behavioural Patterns
Created By	System
Last Updated By	System
Actor	System / Analytics Engine
Description	System identifies declining grades, increasing absenteeism, irregular sleep patterns, and reduced campus engagement using ML-based pattern detection. Runs on a weekly schedule and on demand.
Preconditions	Student data has been collected and is available in the analytics layer.
Postconditions	Pattern analysis report generated; input ready for risk classification (UC-10).
Priority	High
Frequency of Use	High (automated weekly + on-demand per student event)
Normal Course	1. Analytics engine triggered 2. Multi-source data loaded 3. Pattern detection models run 4. Anomalies identified 5. Output passed to UC-10
Alternative Courses	Insufficient data → flag student for manual counsellor review
Exceptions	Model failure; data quality below threshold
Includes	Authentication, Data Validation
Special Requirements	ML accuracy $\geq 85\%$; explainable outputs for counsellors
Assumptions	Sufficient historical data (minimum 4 weeks) available per student
Notes and Issues	Models require periodic retraining as new cohorts are enrolled

UC-10

Use Case ID	UC-10
Use Case Name	Detect Mental Health Risk
Created By	System
Last Updated By	System
Actor	System / Analytics Engine
Description	System classifies each student into Low / Medium / High risk using a weighted algorithm combining PHQ-9, GAD-7, academic delta, attendance, and behavioural pattern outputs from UC-09. Thresholds are configurable by admin.
Preconditions	Pattern analysis (UC-09) has been completed; screening data exists.
Postconditions	Risk tier updated in student profile; alert generated if High threshold crossed (UC-11).
Priority	High
Frequency of Use	High (automated after every UC-09 run and each screening submission)
Normal Course	1. Weighted inputs loaded 2. Scoring algorithm applied 3. Risk tier computed 4. Profile updated 5. If High → trigger UC-11
Alternative Courses	Borderline score → flag for counsellor manual review
Exceptions	Model failure; incomplete input data
Includes	«include» UC-09 (risk detection always requires pattern analysis)
Special Requirements	Single-student classification latency ≤ 5 sec; full batch ≤ 2 hours
Assumptions	Scoring weights have been calibrated to institutional norms by admin
Notes and Issues	Threshold misconfiguration could lead to over/under-alerting; requires governance review

UC-11

Use Case ID	UC-11
Use Case Name	Generate Counsellor Alert
Created By	System
Last Updated By	System
Actor	System / Analytics Engine
Description	System automatically generates and dispatches a real-time alert to the assigned counsellor and relevant administrators whenever a student's composite risk score crosses the High threshold. Alert includes score breakdown, key contributing factors, and a direct link to the student profile.
Preconditions	Risk classification (UC-10) has been completed; student risk tier is High; counsellor assignment exists.
Postconditions	Alert dispatched via in-app notification and email; alert logged with timestamp; counsellor acknowledgement status set to Pending.
Priority	High
Frequency of Use	Medium (event-driven; triggered per High-risk classification)
Normal Course	1. UC-10 triggers High-risk flag 2. System composes alert with score breakdown 3. Alert sent to assigned counsellor via in-app + email 4. Copy sent to administrator 5. Alert logged with timestamp and delivery status
Alternative Courses	No counsellor assigned → alert escalated to admin for manual assignment
Exceptions	Notification service failure; counsellor account inactive or unavailable
Includes	«include» UC-10 (alert always triggered by risk detection)
Special Requirements	Alert delivery latency ≤ 2 minutes from risk classification; retry on failure
Assumptions	Each student has an assigned counsellor at the time of alert generation
Notes and Issues	Repeat alerts for the same student should be suppressed until counsellor acknowledges; de-duplication logic required

UC-12

Use Case ID	UC-12
Use Case Name	Update Dashboards & Generate Reports
Created By	System
Last Updated By	System
Actor	System / Analytics Engine
Description	System refreshes role-specific dashboards for counsellors and administrators and generates automated weekly and monthly institutional mental health trend reports. Counsellor dashboards display prioritised at-risk student lists; admin dashboards display campus-wide KPI analytics.
Preconditions	Risk classification data is available; at least one reporting cycle has elapsed.
Postconditions	Dashboards updated with latest data; scheduled reports generated and made available for export (PDF / CSV).
Priority	High
Frequency of Use	Very High (real-time dashboard refresh + weekly/monthly reports)
Normal Course	1. Scheduled trigger or data update event fires 2. Analytics engine aggregates latest risk, screening, and session data 3. Counsellor dashboard refreshed with sorted at-risk list 4. Admin dashboard refreshed with KPI trends 5. Report generated and stored for download
Alternative Courses	Data unavailable for reporting period → report flagged as incomplete with explanation
Exceptions	Analytics engine failure; database connection error
Includes	Authentication, Data Validation
Special Requirements	Dashboard refresh latency ≤ 30 seconds; reports must be exportable as PDF and CSV
Assumptions	Sufficient data exists for meaningful trend analysis over the reporting period
Notes and Issues	All reports must use aggregated, anonymised data only; no individual student identifiers in admin-level exports

UC-13

Use Case ID	UC-13
Use Case Name	View At-Risk Student List
Created By	System Analyst
Last Updated By	System Analyst
Actor	Counsellor / MH Professional
Description	Counsellor views a prioritised, sortable list of at-risk students assigned to their caseload. Each entry displays the student's current risk tier, composite risk score, key contributing factors (academic delta, screening scores, mood trends), and time since last contact.
Preconditions	Counsellor is logged in; at least one student is assigned to the counsellor's caseload.
Postconditions	At-risk list displayed; counsellor may navigate to individual student profiles from this view.
Priority	High
Frequency of Use	Very High (primary daily workflow for counsellors)
Normal Course	1. Counsellor navigates to dashboard 2. System fetches caseload data filtered by assigned students 3. List rendered sorted by risk tier (High first) 4. Counsellor selects a student to view full profile
Alternative Courses	No students assigned → display empty state with guidance to contact admin
Exceptions	Data fetch failure; analytics engine unavailable
Includes	Authentication
Special Requirements	List must be sortable by risk tier, score, and last-contact date; fast load time
Assumptions	Student-counsellor assignment has been configured by admin
Notes and Issues	Counsellors should only see students assigned to them; cross-counsellor visibility must be restricted by RBAC

UC-14

Use Case ID	UC-14
Use Case Name	Manage Counselling Sessions
Created By	System Analyst
Last Updated By	System Analyst
Actor	Counsellor / MH Professional
Description	Counsellor schedules, reschedules, or cancels counselling sessions with assigned students. The system sends automated 24-hour reminders to both parties and logs all session activity with timestamps.
Preconditions	Counsellor is logged in; student exists in the caseload; session management module is active.
Postconditions	Session record created, updated, or cancelled in the system; notifications sent to student and counsellor; session history updated.
Priority	High
Frequency of Use	High
Normal Course	1. Counsellor selects student from caseload 2. Opens session scheduler 3. Selects date, time, and session type (in-person / virtual) 4. System confirms slot availability 5. Session saved; reminders scheduled automatically
Alternative Courses	Slot conflict detected → system suggests next available time; counsellor reschedules
Exceptions	System failure during save; student account deactivated
Includes	Authentication
Special Requirements	24-hour automated reminder to both parties; session history must be immutable once completed
Assumptions	Counsellor maintains up-to-date availability in the system
Notes and Issues	Cancellations within 2 hours of session time should trigger an escalation notification to admin

UC-15

Use Case ID	UC-15
Use Case Name	Flag Student as Urgent / Escalate
Created By	System Analyst
Last Updated By	System Analyst
Actor	Counsellor / MH Professional
Description	Counsellor manually flags a student as an urgent or critical case requiring immediate escalation beyond standard scheduling workflows. Escalation triggers administrator notification and overrides normal session queue priority.
Preconditions	Counsellor is logged in; student is in the counsellor's caseload; counselling session (UC-14) context exists.
Postconditions	Student flagged as Urgent in the system; admin notified immediately; escalation event logged in audit trail; student fast-tracked in session scheduling.
Priority	High
Frequency of Use	Low (emergency escalation; exception workflow)
Normal Course	1. Counsellor views student profile or session record 2. Selects "Flag as Urgent" action 3. Enters escalation reason (mandatory free-text note) 4. System updates student status to Urgent 5. Admin and welfare committee notified immediately
Alternative Courses	Counsellor cancels flag action → no change to student status
Exceptions	System failure during flag save; admin account unavailable for notification
Includes	«extend» UC-14 (escalation extends standard session management)
Special Requirements	Escalation reason is mandatory; audit trail entry must be tamper-evident
Assumptions	Admin accounts are active and reachable for escalation notifications
Notes and Issues	Escalation flags must be reviewed and resolved within 24 hours; unresolved flags should re-alert admin automatically

UC-16

Use Case ID	UC-16
Use Case Name	Manage User Accounts
Created By	System Analyst
Last Updated By	System Analyst
Actor	Admin
Description	Administrator creates, updates, and deactivates user accounts across all roles (Student, Counsellor, Admin). Account management includes role assignment, counsellor-student caseload mapping, and bulk import of student records at semester start.
Preconditions	Admin is logged in with full administrative privileges; institutional identity source (SIS or LDAP) is accessible.
Postconditions	Account created, modified, or deactivated; user notified of account status change; all actions logged in audit trail.
Priority	High
Frequency of Use	Medium (high at semester start; low during mid-semester)
Normal Course	1. Admin navigates to user management panel 2. Selects action (Create / Update / Deactivate) 3. Enters or modifies user details and role 4. System validates and saves changes 5. User notified via email; action logged
Alternative Courses	Duplicate account detected → system prompts admin to merge or skip; bulk import errors → flagged rows reported in error log
Exceptions	SIS / LDAP connection failure; invalid data format in bulk import
Includes	Authentication, Data Validation
Special Requirements	Bulk import must support CSV with schema validation; deactivation must not delete historical data
Assumptions	Admin has verified identity source records before account creation
Notes and Issues	Deactivated accounts must be soft-deleted to preserve historical session and screening records

UC-17

Use Case ID	UC-17
Use Case Name	Generate Institutional Trend Reports
Created By	System Analyst
Last Updated By	System Analyst
Actor	Admin
Description	Administrator generates aggregated, anonymised mental health trend reports covering campus-wide risk distribution, screening completion rates, counsellor caseload statistics, and intervention outcomes. Reports are exportable as PDF or CSV for welfare committee review and institutional compliance.
Preconditions	Admin is logged in; sufficient data exists for the selected reporting period; analytics engine is available.
Postconditions	Report generated and available for immediate download in selected format; report generation event logged.
Priority	High
Frequency of Use	Medium (weekly / monthly / on-demand)
Normal Course	1. Admin selects reporting period and parameters 2. System aggregates anonymised data from analytics layer 3. Report compiled with charts and summary statistics 4. Admin previews report 5. Admin downloads in PDF or CSV format
Alternative Courses	Insufficient data for selected period → system warns and allows partial report generation
Exceptions	Analytics engine failure; export service unavailable
Includes	Authentication, Data Validation
Special Requirements	All exported data must be fully anonymised; no student-identifiable fields in any export
Assumptions	Anonymisation pipeline has been validated and certified before institutional deployment
Notes and Issues	Reports must comply with applicable data protection regulations (GDPR / FERPA); legal review recommended before first export

UC-18

Use Case ID	UC-18
Use Case Name	Configure System Policies & Settings
Created By	System Analyst
Last Updated By	System Analyst
Actor	Admin
Description	Administrator configures system-wide privacy policies, Role-Based Access Control (RBAC) permissions, data retention periods, risk scoring threshold weights, and notification settings. All configuration changes are version-controlled and logged in the audit trail.
Preconditions	Admin is logged in with configuration privileges; system is in a stable operational state.
Postconditions	Configuration saved and applied system-wide; previous configuration version archived; change event logged in audit trail with admin identity and timestamp.
Priority	High
Frequency of Use	Low (initial setup and periodic policy reviews; not routine)
Normal Course	1. Admin navigates to system configuration panel 2. Selects configuration domain (RBAC / Risk Thresholds / Retention / Notifications) 3. Reviews current settings 4. Applies changes with mandatory confirmation prompt 5. System saves new configuration; archives previous version; logs change
Alternative Courses	Invalid threshold values entered → system rejects with validation error and explanation; admin corrects and resubmits
Exceptions	System failure during save; configuration schema validation error
Includes	Authentication, Data Validation
Special Requirements	All configuration changes require explicit admin confirmation; previous configurations must be restorable from archive
Assumptions	Admin has received institutional authorisation before modifying risk threshold weights or RBAC policies
Notes and Issues	Misconfigured risk thresholds can directly impact student safety; changes to scoring weights should require dual-admin approval as a governance safeguard

Actor Responsibilities

1. Student

- Registers and authenticates using institutional credentials with optional 2FA.
- Completes PHQ-9, GAD-7, and PSS screening questionnaires periodically.
- Logs daily mood, sleep duration, and stressors via the check-in module.
- Views personal mental health status, risk tier, and wellness resources.
- Requests counselling sessions and manages session bookings.

2. System / Analytics Engine

- Collects and integrates multi-source student data weekly from SIS via read-only API.
- Runs ML-based behavioural pattern detection (declining grades, absenteeism, irregular sleep).
- Classifies each student into Low / Medium / High risk using a weighted scoring algorithm.
- Auto-generates counsellor alerts when High threshold is crossed, with contributing factor breakdown.
- Forecasts long-term periodic risk surges using Attention-based Transformer models.
- Updates dashboards and generates automated weekly/monthly institutional reports.

3. Counsellor / Mental Health Professional

- Reviews prioritised at-risk student list sorted by risk tier (High first).
- Accesses full student profiles including screening history, academic data, and session records.
- Schedules, reschedules, and cancels sessions with automated reminder notifications.
- Records structured post-session notes (encrypted; access-controlled).
- Acknowledges and action-logs system-generated alerts to suppress repeat notifications.
- Generates exportable per-student mental health history reports.

4. Administrator

- Creates, updates, and deactivates user accounts across all roles.
- Monitors campus-wide mental health KPI dashboards and trend analytics.
- Configures risk scoring weights, tier thresholds, and data retention policies.
- Assigns students to specific counsellors based on caseload balancing.
- Exports aggregated anonymised trend reports for welfare committee review.

5. Parent / Guardian (Limited Access Actor)

- Views a consent-gated, anonymised well-being summary (no raw clinical data).
- Receives system-generated general wellness notifications if student has granted consent.
- Provides the secondary support layer for at-risk students outside the institution.